DEWDRIVE · MARKET AND VIABILITY

DewDrive market and viability

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	DWD-MKT-001	Version	0.1
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dewdrive		
Source	docs/10-market.md @ ca02742		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First market and viability note (kit 1.8 pipeline); prices checked 2026-10-25

DewDrive market and viability

DewDrive is for arid households, schools and clinics that carry or buy their drinking water, and for the researchers and NGOs who want comparable yield data; it competes with sealed solar hydropanels at about $2,000 a panel and with grid-powered atmospheric water generators. An open design matters because the people who need it cannot buy or repair a sealed product, and because published yields vary so much that only logged, comparable units can show where it is worth building. Prices and benchmarks were checked on 2026-10-25 unless another date is given. Method: STANDARDS section 23.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Arid households	Households carrying or buying water in hot semi-arid regions with night humidity of 25 to 40 % (DWD-PRB-001)	SOURCE solar hydropanel: about $2,000 per panel for about 3 to 5 L per day (Forbes, 2022); sealed, not locally repairable	695 in parts (DWD-CAL-001 J1), plus local labour	About a third of a hydropanel's price, built and repaired locally; about $0.87 per litre over 5 years in parts at 0.44 L per day, against a hydropanel's higher output
Schools and clinics	Sites with staff on hand twice a day and a delivered supply that runs short	Watergen GENNY home atmospheric water generator: up to 20 L per day, 550 W nominal and about 400 Wh per litre at 26.7 °C and 60 % RH; price not listed (Watergen); needs grid power and humid air	695	No grid, works at 25 to 40 % RH where refrigeration-type generators give little
Researchers and NGOs	University field stations and WASH programmes comparing sorbents and sites	No open, logged reference device found; the field station partner type was decided on 2026-10-02	695	Open model, calculation and logger; the sorbent can be swapped for MOF or other composites later

The parts are ordinary: white silica gel beads sell at retail for about €313 per 25 kg drum, VAT included, about €12.50 per kilogram (silica-gel.it), against the $8 per kilogram estimate in the BOM, so the sorbent line may rise by about $15 at small-quantity retail prices; food-grade $CaCl_2$ is a common food additive (E 509). The cost per litre of carried or trucked water in the partner's region is to assess with the partner.

Open route

Local sheet-metal and joinery workshops build the box, trays and stand from the build plan; a coating shop applies the PTFE coatings; electronics distributors supply the panel, battery and logger parts. University field stations run the first units and publish logged yields; water and sanitation NGOs carry the design to schools and households where the data show it is worth the money. CERN-OHL-S-2.0 keeps changes to the hardware open and MIT keeps the scripts open, and the parametric model lets a site change the box size, tilt or sorbent and rerun the numbers.

Sources

- Kart, J. "SOURCE Solar-Powered Hydropanels Can Produce Up to 5 Liters of Drinking Water per Day." *Forbes*, January 13, 2022. https://www.forbes.com/sites/jeffkart/2022/01/13/solar-powered-source-hydropanels-can-produce-up-to-5-liters-of-drinking-water-per-day/ (cited in DWD-PRB-001).
- Watergen. "GENNY Home" product page. https://watergen.com/product-page-genny-home/ (checked 2026-10-25).
- Silica Gel Italia. "25 kg Bulk Silica Gel, White Spherical, Drum." https://www.silica-gel.it/en/spherical-white-silica-gel/153-beaded-white-silica-gel-bulk-drum-20-kg.html (checked 2026-10-25).